

# Module 9 – Big Ideas: Data Collection, Wrangling, EDA & Capstone

## Big Idea 1: Real Data Doesn't Start Clean

In real projects, you are not given a perfect CSV. You have to go get the data yourself.

This module is the first time the workflow looks like a real data job:

- Data lives in APIs, websites, and messy formats
- You have to collect it, shape it, and make it usable

APIs return structured data (JSON), but not in a clean table format

Web scraping pulls data from HTML, which is even messier

Before you can analyze anything, you need to:

- Gather data from multiple sources
- Convert it into a consistent tabular format
- Decide what data actually matters

This is the beginning of the **data pipeline mindset**

Why it matters

Most of the work in data science is not modeling

It is getting data into a usable state

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## Big Idea 2: APIs = How Modern Data Is Accessed

APIs are one of the most important real-world data sources

In this module:

- You send a request to an API
- You get back JSON (nested data, not flat tables)
- You convert it into a DataFrame

Key shift:

You are no longer loading files

You are querying systems

Important concepts:

- Endpoints = different data sources (launches, rockets, etc.)
- JSON = hierarchical data (lists, dictionaries)
- Normalization = flattening JSON into columns

You also see:

- IDs instead of real values (requires additional API calls)
- Multiple joins/enrichment steps to build one dataset

Why it matters

This is how production systems work

You rarely get everything in one query

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## Big Idea 3: Data Wrangling Is Where Projects Are Won or Lost

Once you have the data, it is still not usable

Data wrangling = turning raw data into analysis-ready data

Core tasks:

- Filtering (only Falcon 9 launches)
- Handling missing values (e.g., payload mass)
- Enriching IDs into meaningful features
- Standardizing columns
- Creating a target variable (success vs failure)

Key idea:

You are making decisions about the data

Examples:

- Fill missing payload mass with a mean
- Keep NULL landing pads because they carry meaning
- Convert outcomes into 0/1 for modeling

Why it matters

Bad wrangling = bad model

Clean data = everything downstream works

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## Big Idea 4: EDA = Understanding Before Modeling

Exploratory Data Analysis (EDA) is where you figure out what's actually going on

Before building models, you ask:

- What patterns exist?
- What features matter?
- What relationships are useful?

Examples from this module:

- Success rate improves over time
- Launch site impacts success rate
- Payload mass interacts with success depending on context

Key idea:

**EDA is how you decide what to model**

You also start thinking about:

- Correlation vs causation
- Feature engineering
- Combining variables for better signal

Why it matters

If you skip EDA, your model is guessing

If you do EDA well, your model has a chance

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## Big Idea 5: SQL + Notebooks = Real Workflow

This module introduces a hybrid workflow:

- SQL → aggregation, filtering, grouping
- Python → visualization, modeling, feature engineering

You are:

- Querying data from a database (SQLite)
- Using SQL to summarize patterns
- Using Python to visualize and explore further

Key idea:

Different tools for different jobs

Why it matters

This is exactly how real teams work

No one tool does everything

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## Big Idea 6: The Capstone Is a Business Problem, Not Just Code

This is not just a technical exercise

You are acting as a data scientist solving a real problem:

- Predict if a rocket will land
- Use that to estimate cost and competitiveness

SpaceX is cheaper because of reuse

If you can predict reuse, you can estimate pricing

Key shift:

From:

- “Build a model”

To:

- “Solve a business problem with data”

Why it matters

This is what data science actually is in the real world

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## Big Idea 7: Feature Engineering Starts Here

Even though full modeling comes later, this is where feature thinking begins

You are already:

- Encoding categorical variables (one-hot encoding)
- Creating target variables (0/1 success)
- Combining features (mass + site + time)

Key idea:

**Good features > fancy models**

Why it matters

Most performance gains come from better features, not better algorithms

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## Big Idea 8: Version Control = Reproducibility

This is the first time GitHub shows up as part of the workflow

You are:

- Saving notebooks
- Tracking changes
- Sharing work

Key idea:

Your work should be reproducible and shareable

Why it matters

In real jobs:

- Code lives in repos
  - Notebooks are versioned
  - Work is collaborative
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## Big Idea 9: End-to-End Pipeline Thinking

This module connects everything so far into one flow:

Data Collection → Data Wrangling → EDA → Modeling (next modules)

You are building:

- A dataset from scratch
- Understanding it
- Preparing it for ML

Key idea:

**Data science is a pipeline, not isolated steps**

Why it matters

If one step breaks, everything breaks

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## **Big Idea 10: This Is the Most “Real” Module Yet**

This is the closest thing so far to an actual data science project

You are:

- Working with messy real-world data
- Making judgment calls
- Building something end-to-end

This is where students usually realize:

“I’m actually doing data science now”